Practical Assignment 1 - Exploring Protein Sequence and Structure

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Introduction

For this assignment, I've considered the sequence of Phosphoglycerate kinase (PGK), an important enzyme implicated in Glycolysis. It is well documented that PGK is conserved across all three taxonomic domains, i.e., Archaea, Bacteria, and Eukarya. This report studies sequence similarity and evolutionary distance across PGK homologues. The assignment considers sequence homologues from 10 species, across the three taxonomic domains. Throughout the scripts, for succinctness, I use shortened labels for each species, which are mentioned in the brackets. For Archaea, I've considered *Thermococcus kodakarensis* (theko), *Methanothermus fervidus* (metfv) and *Pyrococcus woesei* (pyrwo). *Escherichia coli* (ecoli), *Geobacillus stearothermophilus* (geose), *Cutibacterium acnes* (cutak) and *Koribacter versatilis* (korve) are considered for Bacteria. The representatives for Eukarya are *Drosophila melanogaster* (drome), *Equus caballus* (eqcab) and *Neurospora crassa* (neucr). The protein sequences were collected through UniProt's programmatic API access (here: https://www.uniprot.org/help/api). Throughout this write-up I will refer to the code written in an Ipython Notebook, as well as other pertinent files that will be shared via Google Drive.

Pairwise Sequence Alignment (PSA)

On using EMBOSS-NEEDLE (Rice et al. 2020, Madeira et al, 2024) for species-wise PSA, and inputting the pertinent sequences in each run on the web interface, the following identity% matrix is received:

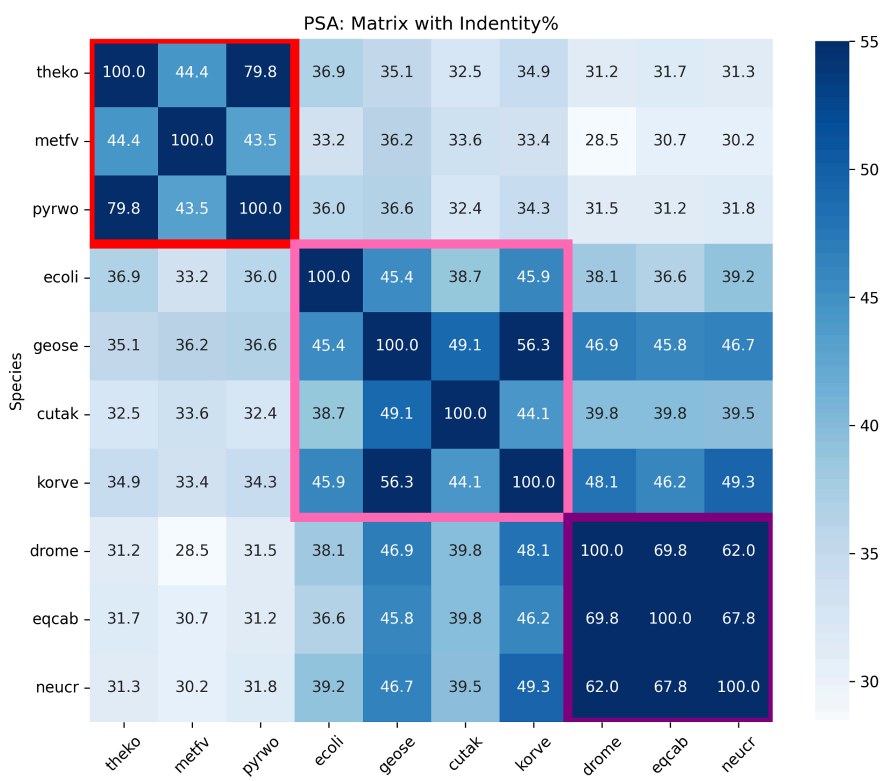


Fig 1: PSA Identity% Matrix. Box outlines represent the domain-domain PSA identity%. Red box is for Archaea-Archaea, Pink is for Bacteria-Bacteria, and Purple is for Eukarya-Eukarya alignments. Notice how intra-domain identity % is higher compared to cross-domain PSAs

Below is an example of a EMBOSS-NEEDLE output:

```
########################################
# Program: needle
# Rundate: Wed 30 Jul 2025 18:50:06
# Commandline: needle
#    -auto
#    -stdout
#    -asequence emboss_needle-I20250730-184957-0160-94143339-p1m.asequence
#    -bsequence emboss_needle-I20250730-184957-0160-94143339-p1m.bsequence
#    -datafile EBLOSUM62
#    -gapopen 10.0
#    -gapextend 0.5
#    -endopen 10.0
#    -endextend 0.5
#    -aformat3 pair
#    -sprotein1
#    -sprotein2
# Align_format: pair
# Report_file: stdout
########################################

#=======================================
#
# Aligned_sequences: 2
# 1: EMBOSS_001
# 2: EMBOSS_001
# Matrix: EBLOSUM62
# Gap_penalty: 10.0
# Extend_penalty: 0.5
#
# Length: 425
# Identity:     141/425 (33.2%)
# Similarity:   213/425 (50.1%)
# Gaps:          53/425 (12.5%)
# Score: 529.0
#
#
#=======================================

EMBOSS_001         1 MFKFYTMDDFDYSGSRVLVRVDINSPVDPHTGRILDDTRMRLHSKTLKEL     50
                      .....|.|.|.:|.||.:|.|:|.||  ..|::..|.|:|....|::..
EMBOSS_001         1 -MSVIKMTDLDLAGKRVFIRADLNVPV--KDGKVTSDARIRASLPTIELA     47

EMBOSS_001        51 VDENAKVAILAHQSRPGKRDFTTMEEHS-----KVLSNILDMPVTYVEDI     95
                     :.:.|||.:.:|..||.:.::.  ||.|     ..|.:.|..||..|:|.
EMBOSS_001        48 LKQGAKVMVTSHLGRPTEGEYN--EEFSLLPVVNYLKDKLSNPVRLVKDY     95

EMBOSS_001        96 FGCAARESIRNMENGDIILLENVRFYSEEVLKRDPKVQAETHLVRKLSSV    145
                     ....      ::..|::::||||||...|  |:|.:.     |.:|.:::
EMBOSS_001        96 LDGV------DVAEGELVVLENVRFNKGE--KKDDET-----LSKKYAAL    132

EMBOSS_001       146 VDYYINDAFAAAHRSQPSLVGF-PLKLPSAAGRLMEREVKTLYKIIKNVE    194
                     .|.::.|||..|||:|.|..|. .....:.||.|:..|:..|.|.:|...
EMBOSS_001       133 CDVFVMDAFGTAHRAQASTHGIGKFADVACAGPLLAAELDALGKALKEPA    182

EMBOSS_001       195 KPCVYILGGVKIDDSIMIMKNILKNGSADYILTSGLVANVFLEASGIDIK    244
                     :|.|.|:||.|:...:.::.::.|  .||.::..|.:||.|:.|.|.|: 
EMBOSS_001       183 RPMVAIVGGSKVSTKLTVLDSLSK--IADQLIVGGGIANTFIAAQGHDV-    229

EMBOSS_001       245 EKNRKILYRKNYKKFIKMAKKLKDKYGEKILTPVDVAINKNGKRI-----    289
                        .|.||..:   .:..||:|....  .|..|.||.:.......     
EMBOSS_001       230 ---GKSLYEAD---LVDEAKRLLTTC--NIPVPSDVRVATEFSETAPATL    271

EMBOSS_001       290 ----DVPIDDIPNFPIYDIGMETIKIYAEKIREAKTIFANGPAGVFEEQQ    335
                         ||..|:    .|.|||..:.:..||.::.||||..|||.||||...
EMBOSS_001       272 KSVNDVKADE----QILDIGDASAQELAEILKNAKTILWNGPVGVFEFPN    317

EMBOSS_001       336 FSIGTEDLLNAIASSNAFSVIAGGHLAAAAEKMGISNKINHISSGGGACI    385
                     |..|||.:.||||.|.|||:..||...||.:..||::||::||:||||.:
EMBOSS_001       318 FRKGTEIVANAIADSEAFSIAGGGDTLAAIDLFGIADKISYISTGGGAFL    367

EMBOSS_001       386 AFLSGEELPAIKVLEEARKRSDKYI    410
                     .|:.|:.|||:.:|||..|:     
EMBOSS_001       368 EFVEGKVLPAVAMLEERAKK-----    387


#---------------------------------------
#---------------------------------------
```

Fig 2: Example output from EMBOSS Needle for the sequence alignment of *Methanothermus fervidus* and *Escherichia coli.* Notice how the identity percent corresponds to the matrix in Fig 1.

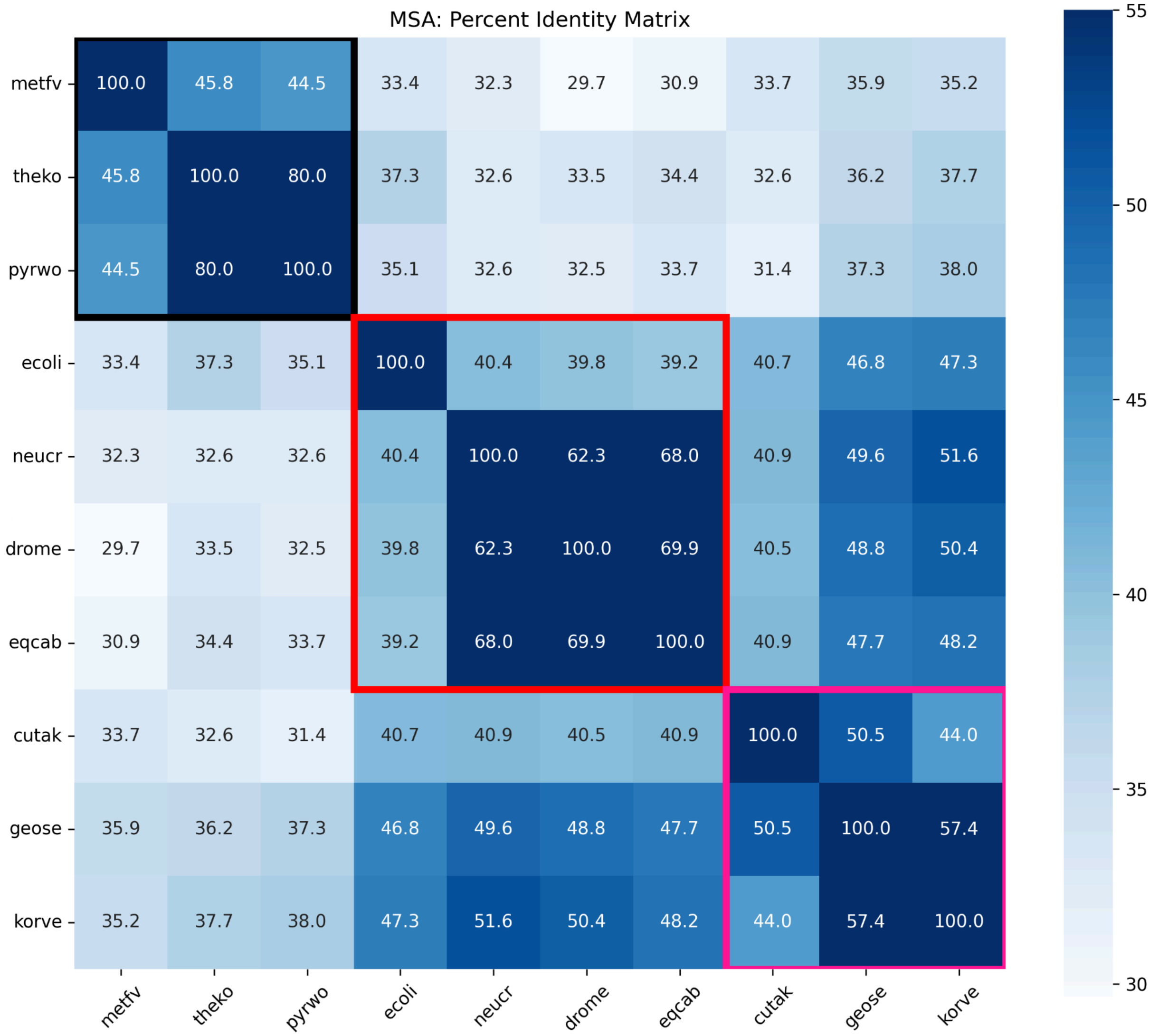
Multiple Sequence Alignment (PSA)

For MSA, I use Clustal Omega (Rice et al. 2020, Madeira et al, 2024), and AlignmentViewer (Reguant et al., 2020) to visualize the alignment, and subsequent conservation analysis. The input to Clustal Omega was a fasta-like text input generated programmatically in the Python Notebook referred. The output of Clustal Omega was a Fasta file, displayed here:

```
>metfv
--MFKFYTMDDFDYSGSRVLVRVDINSPVDPHTGRILDDTRMRLHSKTLKELVDENAKV-
AILAHQSRPGKRDFT---TMEEHSKVLSNILD------MPVTYVEDIFGCAARESIRNME
NGDIILLENVRFYSEEVLKRDP--------KVQAETHLVRKLSSVVDYYINDAFAAAHRS
QPSLVGFPLKLPS-AAGRLMEREVKTLYKIIKNVEKPCVYILGGVKIDDSIMIMKNILKN
GSADYILTSGLVANVFLEAS-GIDIKEKNRKILYRKNYKKFIKMAKKLKDKYGEKILTPV
DVAINKNGKRIDV----PID-----DIPNFPIYDIGMETIKIYAEKIREAKTIFANGPAG
VFEEQQFSIGTEDLLNAIAS---SNAFSVIAGGHLAAAAEKMGISN-KINHISSGGGACI
AFLSGEELPAIKVLEEARKRSDKYI--
>theko
-----MFRITDFTYEGKTVFLRADLNSPVE--KGRITSDARFRAVLPTIQHLLDNGAKL-
VIATHQSRPYKGDYI---TTEEHAQILSRLLG------QEVEYVEDIFGKYARERISSLK
PGEAIILENLRFSAEEVFNRSI--------EECEKTFFVRKLAPLIDYVVNDAFATAHRS
QPSLVGFARLKPM-IMGKLMETEVDALSKAYESEERPRVYVLGGAKVDDSLRVAENVLRK
GKADLILTGGLVGQIFTLAK-GFDLGDANIEFLHRKGLLELVDWAEKILDEFYPYVRTPV
DFAIDYKGERLEV----DLLGGEKRLFDQYPILDIGSRTVEKYREILIGAKIIVANGPMG
VFEREEFAVGTVGVFRAIGE---SPAFSVVGGGHSIASIYRYNIT--GISHISTGGGAML
SFFAGEELPVLKALKISYERFKDRVKE
>pyrwo
-----MFRLRDFEYYNRTVFLRVDLNSPMS--NGKIISDARFRAVLPTIKYLIESGAKV-
VVGTHQGKPYSEEYS---TTEEHARILSELLN------MHVEYVEDIFGKYARERIKAMK
PGEVIVLENLRFSAEEVKNATI--------EECEKTFFVRKLSQVIDLVVNDAFAAAHRS
QPSLVGFARIKPM-IMGFLMEKEVDALTKAYESEEKPRVYVLGGAKVDDSLKVAENVLRK
EKADLILTGGLVGQLFTLAK-GFDLGRENIKFLEKKGILKYVDWAEKILDEFYPYVRTPV
DFAIDFKGERVEI----DLLSDEKRLFDEYPILDIGSRTVEKYREILLKARIIVANGPMG
VFEREEFAVGTIGVFKAIGE---SPAFSVIGGGHSIASIYKYNIT--GISHISTGGGAML
TFFAGEKLPVLEALKISYEKFSNLLS-
>ecoli
---MSVIKMTDLDLAGKRVFIRADLNVPVK--DGKVTSDARIRASLPTIELALKQGAKV-
MVTSHLGRPTEGEYNEEFSLLPVVNYLKDKLSNPVRLVKDYL------------DGVDVA
EGELVVLENVRFNKGEKKD-----------D----ETLSKKYAALCDVFVMDAFGTAHRA
QASTHGIGKFADVACAGPLLAAELDALGKALKEPARPMVAIVGGSKVSTKLTVLDSLSKI
--ADQLIVGGGIANTFIAAQ-GHDVGKSLYEADLVDEAKRLLTT---------CNIPVPS
DVRVATEFSETAPA----TLKSVNDVKADEQILDIGDASAQELAEILKNAKTILWNGPVG
VFEFPNFRKGTEIVANAIAD---SEAFSIAGGGDTLAAIDLFGIAD-KISYISTGGGAFL
EFVEGKVLPAVAMLEERAKK-------
>neucr
MSLSNKLSIEDVDLKGKRVLIRVDFNVPLDA-EKKVTNPQRIAGAIPTIKYALDHGAKAV
VLMSHLGRPDGKP-NPKYSLKPVVPELEKLLG------KKVTFAPDCVGPEVEEIVNKAD
NGEVILLENLRFHIEEEGKGTDAEGNKVKADKAKVEEFRKNLTKLGDVYINDAFGTAHRA
HSSMVGID--LPVKAAGFLMKKELQYFAKVLESPQRPFLSILGGAKVSDKIQLIDNLLDK
--VNTLIVCGGMAFTFKKTLQNMPIGNSLFDEAGAKIVPDLVKKAEK----NNVKLVLPV
DFTIADKFDKDANT---GYATDKDGIPDGWMGLDCGEESVKLFTQAINESQTILWNGPAG
VFEFDKFAKGTKATLDACVKAAEEGRTVIIGGGDTATVAAKYGVED-KLSHVSTGGGASL
ELLEGKALPGVVALSERQ---------
>drome
-MAFNKLSIENLDLAGKRVLMRVDFNVPIKE--GKITSNQRIVAALDSIKLALSKKAKSV
VLMSHLGRPDG-NKNIKYTLAPVAAELKTLLG------QDVIFLSDCVGSEVEAACKDPA
PGSVILLENVRFYVEEEGKGLDASGGKVKADPAKVKEFRASLAKLGDVYVNDAFGTAHRA
HSSMMGDG--FEQRAAGLLLNKELKYFSQALDKPPNPFLAILGGAKVADKIQLIENLLDK
--VNEMIIGGGMAFTFLKVLNNMKIGGSLFDEEGSKIVEKLVEKAKK----NNVQLHLPV
DFVCGDKFAENAAV---SEATVEAGIPDGHMGLDVGPKTRELFAAPIARAKLIVWNGPPG
VFEFPNFANGTKSIMDGVVAATKNGTVSIIGGGDTASCCAKWNTEA-LVSHVSTGGGASL
ELLEGKTLPGVAALTSA----------
>eqcab
MSLSNKLTLDKLNVKGKRVVMRVDFNVPMKN--NQITNNQRIKAAVPSIKFCLDNGAKSV
VLMSHLGRPDVGPMPDKYSLQPVAVELKSLLG------KDVLFLKDCVGPEVEKACADPA
AGSVILLENLRFHVEEEGKGKDASGNKVKAEPAKIETFRASLSKLGDVYVNDAFGTAHRA
HSSMVGVN--LPQKAGGFLMKKELNYFAKALESPERPFLAILGGAKVADKIQLINNMLDK
--VNEMIIGGGMAFTFLKVLNNMEIGTSLFDEEGAKIVKNLMSKAEK----NGVKITLPV
DFVTADKFDENAKT---GQATVASGIPAGWMGLDCGTESSKKYAEAVARAKQIVWNGPVG
VFEWEAFARGTKALMDEVVKATSRGCITIIGGGDTATCCAKWNTED-KVSHVSTGGGASL
ELLEGKVLPGVDALSNV----------
>cutak
----MKSVEDLGDLRGKRVVIRCDFNVPLD--GDKITDDGRIKAALPTLTQLHEAGART-
TVLAHLGRPKGAV-DPKYSLAPVAKRLGELMNLDVKL------AGDVVGPSATKTSESLA
DGEIALVENVRFEKGETSK-----------DEAERAELAQKYASFGEFYVSNGFGVVHRK
QASVYDVAKLLPS-AAGSLVAKEVEVLEGLTADPKRPFVVVLGGAKVADKLAVIDNLLKA
--ADTLVIGGGMAYTFLKAK-GLEVGDSLLDEDSIDACTKYLEQAKA----AGKQILLPV
DVRVVSDIDFEARTVGQIDVVPADEIPSDKMAVDIGPETEKAYANAIKGAKSVFWNGPMG
VFEIKGLELGTKAVAQALTE--V-DGLSVVGGGDSASAVRTLGFADDQFGHISTGGGASL
ELMEGKELPGIAVLKENN---------
>geose
---MNKKTIRDVDVRGKRVFCRVDFNVPME--QGAITDDTRIRAALPTIRYLIEHGAKV-
ILASHLGRPKGKV-VEELRLDAVAKRLGELLERPVAK------TNEAVGDEVKAAVDRLN
EGDVLLLENVRFYPGEEKN-----------D----PELAKAFAELADLYVNDAFGAAHRA
HASTEGIAHYLPA-VAGFLMEKELEVLGKALSNPDRPFTAIIGGAKVKDKIGVIDNLLEK
--VDNLIIGGGLAYTFVKAL-GHDVGKSLLEEDKIELAKSFMEKAKE----KGVRFYMPV
DVVVADRFANDANT---K-VVPIDAIPADWSALDIGPKTRELYRDVIRESKLVVWNGPMG
VFEMDAFAHGTKAIAEALAE--ALDTYSVIGGGDSAAAVEKFGLAD-KMDHISTGGGASL
EFMEGKQLPGVVALEDK----------
>korve
---MSKLSIKDLQLSNKRVFMRVDFNVPLDE-NGRVTDDTRIRETLPTIEYALRHGAKL-
ILCSHLGRPKGKP-NPKMSLKPVAERLRVMLDHAISPGQNVGFSPDCIGMQAQEMAKQLE
KGQALLLENVRFHAEEEKN-----------D----PAFAKELASLCELYVNDAFGSAHRA
HASTEGITHYVEKSAAGLLMQKELDYLGKATSNPAKPFVAILGGAKVSDKIGVIQNLMAK
--VDAIIIGGGMAYTFLKAQ-GQEIGKSLFEADKLDLAKQILADAHK----RGLKFLLPV
DHVTADKFDMHATP---HQIGEGQSIPAEQMALDIGPKTVALFSEEIAKARTIVWNGPMG
VFEFDNFAKGTRAIAKAVAG--NSGATSIVGGGDSVAAVHDAGVAD-KITHISTGGGASL
EFLEGKKLPGVEALTNK----------
```

Fig 3: Clustal Fasta Output on Multiple Sequence Alignment

We can observe a similar identity matrix when we consider the MSA Percent Identity matrix obtained from Clustal.



	metfv	theko	pyrwo	ecoli	neucr	drome	eqcab	cutak	geose	korve
metfv	100.0	45.8	44.5	33.4	32.3	29.7	30.9	33.7	35.9	35.2
theko	45.8	100.0	80.0	37.3	32.6	33.5	34.4	32.6	36.2	37.7
pyrwo	44.5	80.0	100.0	35.1	32.6	32.5	33.7	31.4	37.3	38.0
ecoli	33.4	37.3	35.1	100.0	40.4	39.8	39.2	40.7	46.8	47.3
neucr	32.3	32.6	32.6	40.4	100.0	62.3	68.0	40.9	49.6	51.6
drome	29.7	33.5	32.5	39.8	62.3	100.0	69.9	40.5	48.8	50.4
eqcab	30.9	34.4	33.7	39.2	68.0	69.9	100.0	40.9	47.7	48.2
cutak	33.7	32.6	31.4	40.7	40.9	40.5	40.9	100.0	50.5	44.0
geose	35.9	36.2	37.3	46.8	49.6	48.8	47.7	50.5	100.0	57.4
korve	35.2	37.7	38.0	47.3	51.6	50.4	48.2	44.0	57.4	100.0

Fig 4: MSA Identity% Matrix. Box outlines follow the same schema as in Fig 1.

The identity % continues to be higher in these boxes compared to cross-domain MSA identity %, consistent with the observations in Fig 1. To further validate this, we can compare PGK sequence identity between species from the same domain and those from different domains, as in the following figure:

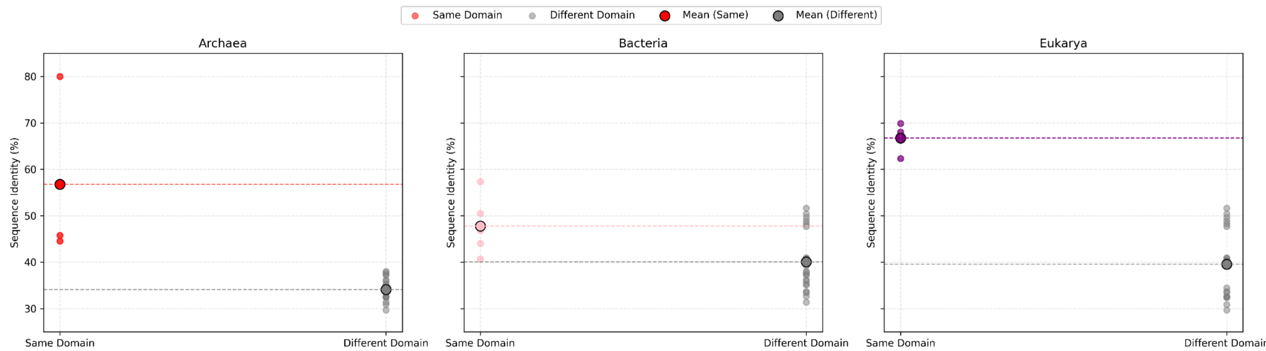


Fig 5: Dot plot for sequence identity within and across domains. Colored dots show intra-domain comparisons; gray dots show cross-domain comparisons. Dashed lines indicate domain means.

We can see that each domain shows markedly higher intra-domain identity than compared to cross-domain identity. This reinforces the idea of evolutionary conservation within taxonomic groups.

Also, it is noticed that there isn't a significant difference between the Matrices in Fig 1 and Fig 4, as shown by this difference heatmap here:

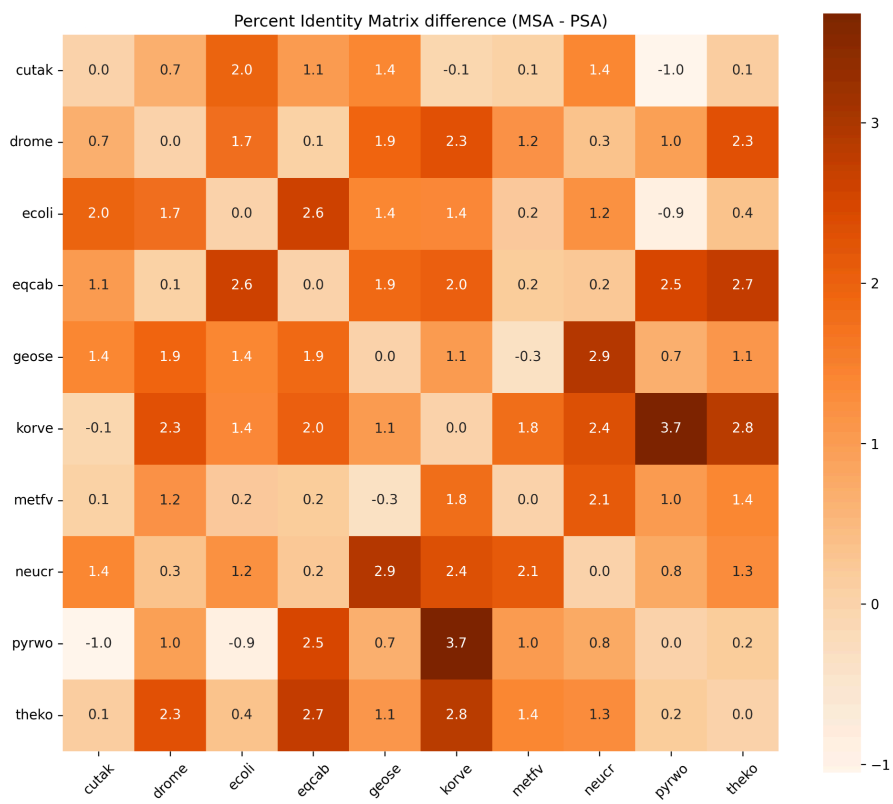


Fig 6: MSA - PSA difference matrix shows no great differences in MSA and PSA results.

Conservation Analysis

For conservation analysis, I use AlignmentViewer to highlight the residues conserved in >= 70% of the sequences (i.e., present in at least 7 out of 10 aligned sequences). Thresholds lower than 70% had a lot of noise. This is the order of the sequences according to which AlignmentViewer visualises the alignments: metfv, theko, pyrwo, ecoli, neucr, drom, eqcab, cutak, geose and korve. The figure below stitches different parts of the visualisation to give a comprehensive view of all the conserved sequences.

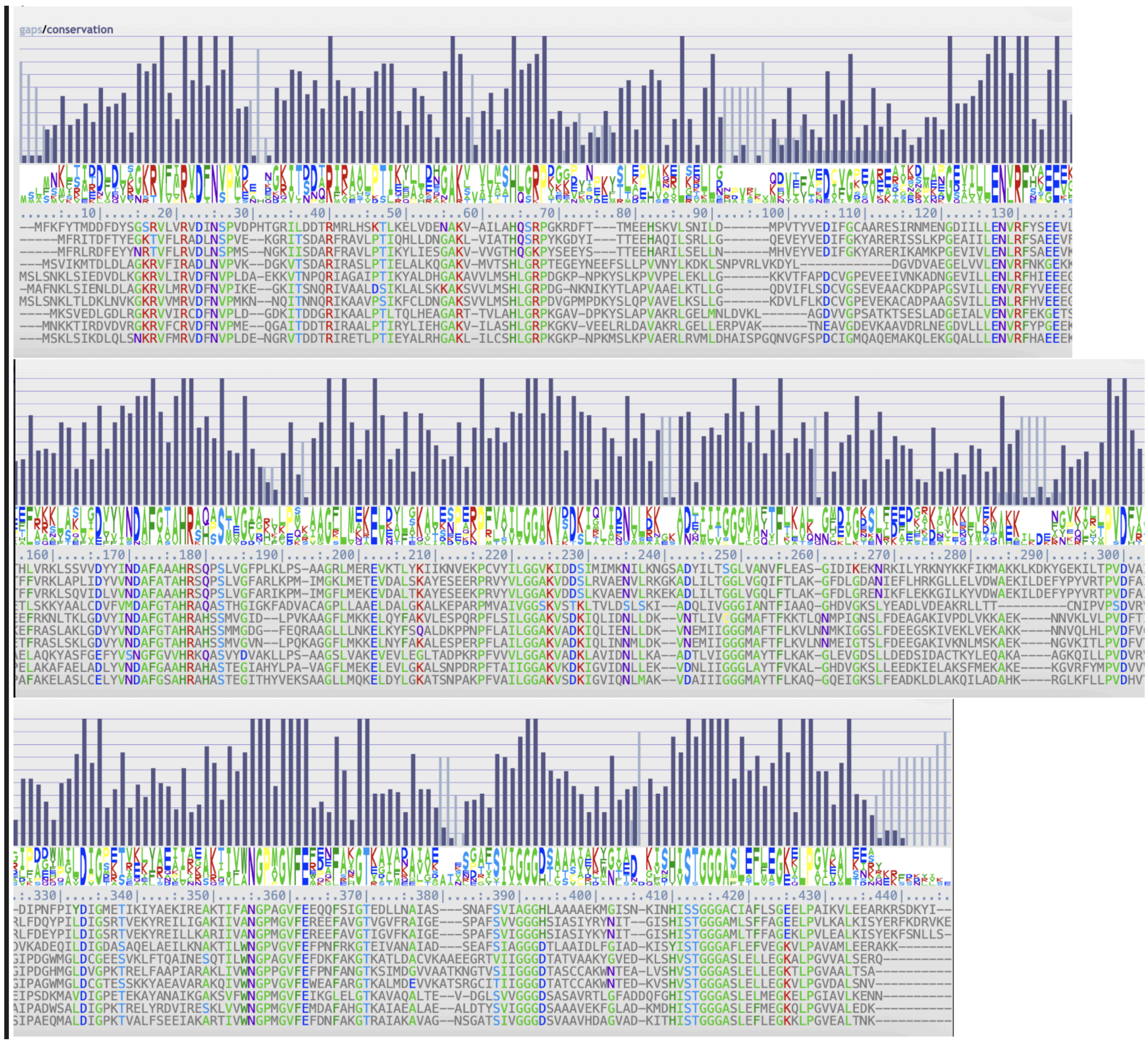


Fig 7: AlignmentViewer output. This visualisation shows only those segments of the sequence that contained conserved residues.

As a mentioned aim of this analysis was to understand the relationship between sequence similarity and evolutionary distance, I used EMBOSS distmat (Rice et al. 2020, Madeira et al, 2024) to construct distance matrices, choosing the Jukes Cantor correction, inputting the MSA Fasta file. This was the output:

OUTPUT FILE outfile

```
Distance Matrix
---------------

Uncorrected for Multiple Substitutions
Gap weighting is 0.000000

            1       2       3       4       5       6       7       8       9      10
         0.00   87.80   20.00   92.76   92.64   82.09   88.83   92.94   94.40   94.16          theko 1
                 0.00   87.80   91.99   92.64   92.79   82.63   94.63   93.66   92.93          metfv 2
                         0.00   91.99   92.64   83.08   88.83   91.95   94.88   94.39          pyrwo 3
                                 0.00   85.01   94.32   89.41   90.18   92.51   92.76          ecoli 4
                                         0.00   95.18   89.34   94.16   91.37   92.64          geose 5
                                                 0.00   92.54   93.03   93.03   92.79          cutak 6
                                                         0.00   94.04   92.31   92.56          korve 7
                                                                 0.00   93.49   95.42          drome 8
                                                                         0.00   37.41          eqcab 9
                                                                                 0.00          neucr 10
```

Fig 8: Output of Emboss-Distmat showing the evolutionary distance matrix.

On calculating the Pearson correlations between each species' pairwise distances (from the distance matrix) and its corresponding % identity values from the MSA (excluding self-comparisons), we can construct this table:

	Domain	Species	Correlation (r)
1	Archaea	theko	-0.993
2	Archaea	metfv	-0.998
3	Archaea	pyrwo	-0.992
4	Bacteria	ecoli	-0.998
5	Bacteria	geose	-0.997
6	Bacteria	cutak	-0.996
7	Bacteria	korve	-0.997
8	Eukarya	drome	-0.991
9	Eukarya	eqcab	-0.992
10	Eukarya	neucr	-0.993

Fig 9: Correlation between pairwise evolutionary distance and MSA identity (%).

The results show strong negative correlations ($r \approx -0.99$) for all species, across all domains. This highlights that generally, as sequence identity increases, evolutionary distance decreases.

I also used EMBL-EBI's Simple Phylogeny tool (Madeira et al, 2024) to construct a phylogenetic tree from the MSA fasta file. Default parameters and heuristics were used to obtain the following tree:

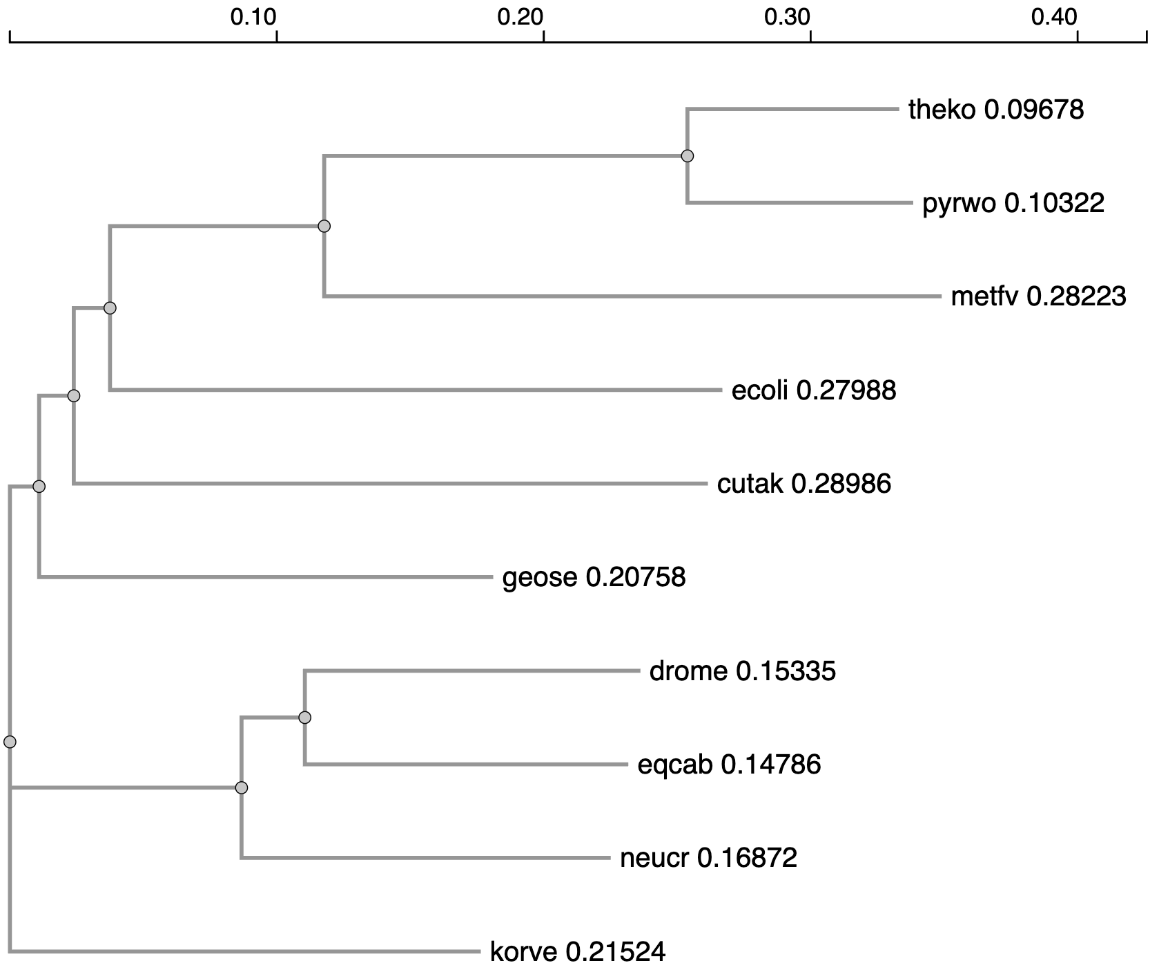


Fig 10: Phylogram generated by Simple Phylogeny. The numbers on the image represent a measure of inferred phylogenetic distance.

The phylogenetic tree clusters species from Archaea, Bacteria, and Eukarya based on PGK sequence. While most taxa group according to their domains, *Koribacter versatilis* (korve) appears as an outgroup. A possible reason is that despite being bacterial, *K. versatilis* belongs to a deeply branching phylum. Its placement likely reflects genuine phylogenetic distance from other bacterial representatives rather than misclassification. It is also worth noting that this topology is based on a single protein sequence, and thus observed divergence reflects protein-level variation and may not directly correspond to taxonomic divergence.

Mapping Conserved Residues onto 3D protein structure

For this part of the assignment, I consider the *E. coli* PGK protein structure, as the crystal structure was readily available on RSCB/PDB (Marqusee, 2005). I annotated all sites in RSCB's Mol* (Sehnal et al. 2021) that were conserved in >70% of the sequences considered in pink, based on the AlignmentViewer output. This is visible in this figure:

Fig 11: *E. coli* PGK crystal structure (X-ray diffraction) annotated with sites conserved in >=70% of the sequences

Out of the 387 amino acids in *E. coli* PGK, 117 residues were conserved in ≥70% of the sequences analyzed, corresponding to ~30% overall conservation. Of these, 70 were located within α-helices or β-sheets, yielding a ratio of 70:47 for conserved residues in structured elements versus loops. This implies a nearly 1.5-fold enrichment of conserved residues in secondary structural regions compared to unstructured regions.

This pattern of enrichment is consistent with expectations, as α-helices and β-sheets generally constitute the structural core of proteins and play a critical role in maintaining the functionality of the enzyme. The catalytic mechanism of PGK involves a large conformational transition (Shirakihara & Evans, 1988), meaning that structural integrity within protein domains is therefore likely essential for this dynamic mechanism. These constraints, especially pronounced in enzymes like PGK that are essential to central metabolism, subject such residues to strong selection pressure. Consequently, we can infer that conserved residues within α-helices and β-sheets are hence associated with functional roles.

References

Marqusee, S. (2005). Crystal structure of the Escherichia coli phosphoglycerate kinase (PGK) [PDB ID 1ZMR]. RCSB Protein Data Bank. https://doi.org/10.2210/pdb1ZMR/pdb

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Shirakihara, Y., & Evans, P. R. (1988). Crystal structure of the complex of Escherichia coli phosphoglycerate kinase with ADP: Implications for substrate-induced conformational change. Journal of Molecular Biology, 204(4), 973–991.